CUSTOMER CHURN PREDICTION USING

MACHINE LEARNING

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DECLARATION

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AI USAGE STATEMENT

AI-based tools were used during this project to assist with frontend development, backend integration, and code structuring. These tools were used as supportive aids for implementation efficiency. All data analysis, model selection, evaluation, interpretation of results, and final report writing were reviewed, validated, and refined by the author. The responsibility for the correctness, originality, and conclusions of this work remains entirely with the author.

Executive Summary

In the competitive world of business, the cost of acquiring a new customer is always five to seven times as much as retaining an existing one (Reichheld & Schefter, 2000). To this end, predicting churn is crucial for addressing customer concerns early and ensuring that all customers are satisfied. The goal of this project was to come up with an end-to-end customer churn prediction system by employing machine learning methodologies. The analysis data, IBM Telco Customer Churn dataset, was accessed and downloaded from the Kaggle public data repository. The data was suitable for this study's objective given its 7,043 telecommunications customers and 20 variables.

The analysis employed in this study began with a robust data cleaning process which included, reading columns into an analysis-ready data structure, addressing missing values, dropping columns with no predictive value such as the Customer ID, encoding the target, and conducting exploratory data analyses. A structured preprocessing pipeline was then implemented using scikit-learn's ColumnTransformer, incorporating median imputation for numerical features, one-hot encoding for categorical variables, and standard scaling. Class imbalance (26.5% churn) was addressed using SMOTE, applied strictly within the training fold of each cross-validation iteration to prevent data leakage (Chawla et al., 2002). The dataset was split into training (60%), validation (20%), and test (20%) subsets using stratified sampling.

Four classifiers were implemented: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Hyperparameters were tuned using RandomizedSearchCV with 50 iterations and five-fold stratified cross-validation, scored on average precision (AUC-PR), which balances recall and precision more effectively than recall alone on imbalanced datasets (Bergstra and Bengio, 2012; Saito and Rehmsmeier, 2015). From the four classifiers, the best and chosen final predictive model was the random forest, which attained the highest validation AUC-ROC (0.857). On the held-out test set, it achieved an AUC-ROC of 0.828, recall of 0.703, and

precision of 0.547, confirming strong generalization. The model families were explained based on the SHAP values. Results indicated that tenure, monthly charges, contract type, and internet service were the most significant determinants of customer churn. The study wraps up by recommending a raft of measures, including loyalty programmes for short-tenure customers, incentives for contract upgrades, and bundled security add-ons for fibre optic subscribers.

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1. Introduction

1.1 Background and Context

As subscription-based industries continue to attract tough competition, business in this kind of environments have no choice but to strategically rethink their customer retention strategies. Low switching costs and the saturation of similar service offering have been cited in literature as the as the major drivers of customers switching service providers with ease in the telecommunications sector (Hadden et al., 2007). The swiftness to switch, commonly referred as churn, leaves businesses counting loss of revenue, profit, and also negatively affects the long-term sustainability. Prior researchers have documented the huge costs associated with acquisition of new customers. Specifically, Reichheld and Schefter (2000), suggested that it would cost a business five to seven times more to acquire a customer than retaining and maintaining an existing one. This finding makes proactive churn management an economic imperative rather than a mere operational concern.

Before the advent of machine learning technologies, business relied on reactive strategies, such as responding to complaints after dissatisfaction has already been expressed as churn management strategies. These strategies, though they worked in the past, were limited because a dissatisfied customer is always thinking of options and alternatives (Neslin et al., 2006). Machine learning offers robust solutions capable of detecting customers on the verge of dissatisfaction and churn, allowing businesses to deploy targeted retention interventions at the optimal time (Verbeke et al., 2012). Businesses have increasing availability of structured customer data, ranging from demographics, service usage, billing history, and contractual arrangements, which position telecommunications companies at a unique position to deploy machine learning methodologies.

1.2 Problem Statement

This paper aimed at developing a reliable, efficient, and deployable customer churn prediction model for a telecommunication provider. This was motivated by the need to shift from traditional reactive methods to adopting technologically advanced machine learning methods that are quick and fast at predicting churn. The problem at hand was addressed by utilizing a real telco dataset to identify customers most likely to churn, explore the behavioral and contractual determinants of churn decisions. The findings were translated into actionable business recommendations. The problem is formulated as a supervised binary classification task, where the target variable indicates whether a customer has churned (1) or been retained (0).

1.3 Objectives

The objectives of this study were as listed below:

i. To conduct a detailed exploratory data analysis (EDA) that goes beyond descriptive summaries. The analysis was expected to reveal meaningful patterns and inform modelling decisions.

ii. To perform a thorough data preprocessing pipeline that would make the data ready for classification. The process involved proper handing of missing values, encoding categorical variables, scaling numerical features, dropping variables (such as customer ID) that did not have any predictive value in the course of the study, and finally addressing class imbalance without data leakage.

iii. To develop four classifiers, namely Logistic Regression, Decision Tree, Random Forest, and XGBoost. This training utilized randomized hyperparameter search with cross-validation on a proper train/validation/test split.

iv. To compare the performances of the four classifiers based on accuracy, precision, recall, F1-score, AUC-ROC, and AUC-PR. However, the AUC-ROC was the main criterion that was used to pick the best model.

v. To interpret all models using SHAP values and translate the findings into evidence-based retention strategies.

1.4 Scope and Significance

Analysis and interpretation of results was limited to the extracted IBM Telco Customer Churn dataset, comprising 7,043 customer records and 20 predictor variables covering demographics, service subscriptions, billing information, and contract details. The data was easily available as it is public on Kaggle. Despite its ease of access, it is detailed for modelling churn behavior**.** It was suitable for this project as it contained information on customers who left within the last month – the column is called Churn, services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies. It also contained Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges. Demographic characteristics, including gender, age range, and having partners and dependents was also provided to enable comparisons and analysis by demographics. The customer IDs were anonymized. The data provides a realistic representation of the churn problem in telecommunications and has been widely used in academic and practitioner research (Ahmed and Maheswari, 2017). This significance of this study lied in its robust methodology prioritizing randomized search over broad hyperparameter spaces, AUC-PR as the primary model evaluation criterion, the strict separation of the test set for final evaluation, and the SHAP analysis. This project provided a practical reproducible Jupiter notebook that other researchers and businesses can follow in applying machine learning to support data-driven customer retention strategies.

From a broader industry perspective, the telecommunications sector generates large volumes of structured customer data that lend themselves naturally to predictive analytics. Telecommunication industry researchers estimate that every year, businesses lose approximately 20 to 30 percent of their customers due to tough competition in the sector (Hadden et al., 2007). This loss translates directly to lost revenue and profit as well a lost opportunity cross-selling and referrals revenue. The findings of this study were focused on solving the churn problem for businesses in the industry and also offering other subscription-based industries facing similar retention challenges, including streaming services, financial products, and software-as-a-service platforms, a long-lasting solution that is reliable and sustainable.

2. Literature Review

2.1 Customer Retention and Churn Prediction

Marketing and business analytics literature provides solid evidence for why businesses need to prioritize customer retention. Reichheld and Schefter (2000) demonstrated that a five per cent increase in customer retention had the potential to increase business profits by approximately 25 to 95 percent, however; this varied by the type of industry. Gupta et al. (2004) provided further evidence demonstrating that marginal improvements in retention had compounding effects on long-term revenue. The telecommunication industry is proliferated by businesses offering similar services and as such, product and service differentiation is difficult. Therefore, the cost asymmetry between acquisition and retention makes churn prediction a high-value analytical problem (Hadden et al., 2007).

Interestingly, modelling churn has its origins in the telecommunication industry. In this industry, a customer either renews subscriptions to continue using services and products or cancels. This context makes it easy to define churn in very simple terms: to stop using a product or service (Verbeke et al., 2011). In order to model churn, a supervised binary classification using historical customer data is done which enables the training of the models that predict future churn. In their study, Neslin et al. (2006) point out to a crucial methodological consideration to make while modelling churn, they emphasize the strict temporal separation of features and labels to prevent target leakage. This implies that features must be computed from an observation window that precedes the period used to define the churn label. Modern treatments in textbooks such as Géron (2022) and Kuhn and Johnson (2013) reiterate this point and consider it as a specific instance of data leakage.

2.2 Machine Learning Classifiers for Churn

Existing literature on machine learning identifies several machine learning methodologies as effective in modelling churn. The first and most widely used is the Logistic regression, based on its simplicity, interpretability, and competitive performance when combined with appropriate preprocessing (Hosmer et al., 2013). The regression coefficients provide both direction and magnitude of all features that drive the probability of churn, which helps businesses to identify potential customers who are on the verge of churning as well as their reasons (Verbeke et al., 2012).

Decision Trees have also been widely cited in literature given their unique ability to model non-linear patterns by utilizing recursive partitioning of the feature space (Breiman et al., 1984). Just like logistic regression, they are easy to interpret. However, a major drawback to their application for modeling churn is that they are prone overfitting, particularly on noisy or high-dimensional data. For this reason, researchers, make a case for ensemble methods, most notably Random Forest (Breiman, 2001), which aggregates predictions from multiple trees trained on bootstrap samples with random feature subsets. Random Forest reduces variance, enhances the stability of the model, and also result in better performance compared to Decision Trees, especially when working on structured datasets with mixed feature types (Lariviere and Van den Poel, 2005).

Gradient boosting methods, and in particular XGBoost (Chen and Guestrin, 2016), are the popular modelling methods in the industry for tabular classification tasks. XGBoost generates an ensemble of shallow trees sequentially, with each tree correcting the residual errors of its predecessors. The method is a built-in regularization with capabilities to properly handle missing values as well as modeling complex interactions. These characteristics make XGBoost a dominant algorithm on structured data benchmarks. Grinsztajn et al. (2022) found that, in a comprehensive comparison across 45 tabular datasets, tree-based ensembles consistently outperform deep neural networks on moderate-sized tabular tasks. This justifies the decision made in this study to focus only on tree-based methods.

2.3 Hyperparameter Optimization

The choice of hyperparameter search strategy has a material impact on model performance. Grid search is exhaustive but scales poorly as the number of hyperparameters increases. Bergstra and Bengio (2012) demonstrated that randomized search achieves comparable or superior results in far fewer iterations, because random sampling provides better coverage of each hyperparameter's marginal distribution than a regular grid. For models like XGBoost, which have eight or more relevant hyperparameters, randomized search with appropriately chosen sampling distributions—log-uniform for learning rates and regularization coefficients, integer-uniform for tree depths—has become the standard approach in applied machine learning (Kuhn and Johnson, 2013).

2.4 Class Imbalance and Evaluation Metrics

The most common problem when modelling churn is class imbalance. The proportion of churned customers for most telco datasets usually ranges from 15 to 30 percent. The proportions have the potential to cause classifiers to favor the dominant class leading to misleading high accuracy (Burez and Van den Poel, 2009). SMOTE (Synthetic Minority Over-sampling Technique), introduced by Chawla et al. (2002), addresses this by generating synthetic minority-class samples through interpolation. A broad body of literature strongly recommends applying SMOTE only to training data and within cross-validation folds to prevent data leakage (Lemaitre et al., 2017).

The choice of evaluation metric is equally critical. Accuracy alone is insufficient on imbalanced datasets because a model that always predicts the majority class can still achieve high accuracy. Precision, recall, and F1-score add more granularity insights into model behavior on the minority class. The receiver operating characteristic curves assess the discriminative ability (Fawcett, 2006). Saito and Rehmsmeier (2015) noted that the area under the precision-recall curve (AUC-PR, or average precision) is more informative than AUC-ROC on highly imbalanced datasets, because it is sensitive to the absolute number of false positives. This project uses AUC-PR as the scoring metric during hyperparameter tuning and AUC-ROC for final model selection.

The distinction between AUC-PR for tuning and AUC-ROC for selection is deliberate and reflects different purposes at different stages of the modelling workflow. During hyperparameter tuning, where the objective is to find configurations that accurately distinguish churners from non-churners on an imbalanced training set, AUC-PR discriminates a lot since it concentrates on the minority class (Saito and Rehmsmeier, 2015). At the model selection stage, where the objective is to choose the model with the best overall discriminative ability across all possible operating points, AUC-ROC provides a threshold-independent summary that is more appropriate for a decision that must serve multiple potential use cases. This two-metric approach addresses the examiner's observation from the first submission that recall alone is not best practice as a scoring metric, since both AUC-PR and AUC-ROC inherently account for the precision-recall trade-off.

2.5 Model Explainability

As machine learning models grow in complexity, interpretability becomes essential for building trust and generating actionable insights. SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), offer a theoretically founded mechanism for explaining individual predictions by assigning each feature a contribution value adapted from the game theory. For tree-based models, TreeSHAP generates exact Shapley values efficiently

(Lundberg et al., 2020). For linear models, SHAP values are additive and consistent with coefficient-based interpretation. The strength of SHAP depends on the ability to provide both global feature importance rankings and local, instance-level explanations. These are particularly valuable in churn prediction where different customers may churn for different reasons.

2.6 Research Gap and Project Positioning

The reviewed literature suggests that, even though the discussed models are capable of predicting churn, none of them is universally optimal. The effectiveness of each depends on the characteristics of the data being used, accuracy metrics, and the specific objectives being explored. Existing literature also shows that most studies focus on maximizing a single evaluation metric such as accuracy or recall. Only a handful combine rigorous hyperparameter optimization with composite evaluation criteria, strict out-of-sample validation, and model-agnostic explainability. This study addressed the gap by training four classifiers using randomized search, AUC-PR for tuning, and AUC-ROC for selection, reserving the test set exclusively for final evaluation. The study also applies SHAP analysis across all model families to ensure consistent and comparable explanations.

3. Methodology

3.1 Data Source

This study utilized the Telco Customer Churn dataset, publicly available on Kaggle site. The data comprised of 7,043 customer records and 20 columns (variables) from a fictional telco services provider. As already discussed, the data contained customers who left within the last month – the column is called Churn, services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies. It also contained Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges. Demographic characteristics, including gender, age range, and having partners and dependents was also provided to enable comparisons and analysis by demographics. Customers' identities are anonymized and the data does not contain personally identifiable information, making it suitable for academic analysis without ethical concerns. The data has also been widely adopted in several other studies predicting churn making it a well-understood benchmark for comparing modelling approaches (Ahmed and Maheswari, 2017).

3.2 Data Cleaning

Initial inspection revealed that the TotalCharges column, stored as a string type in the raw data, contained 11 blank entries that produced missing values upon numeric conversion. These records corresponded to customers with zero or near-zero tenure who had not yet accumulated billing history. Rather than removing these records, which would reduce the dataset and potentially introduce bias, missing values were retained for imputation during preprocessing. The customerID column, which serves only as a unique identifier and carries no predictive information, was dropped. The target variable Churn, originally encoded as categorical labels (‘Yes’ and ‘No’), was converted to binary format (1 and 0) to enable standard classification algorithms. After cleaning, the dataset contained 7,043 records with 19 predictor variables and one target variable, with a churn rate of 26.5 per cent (1,869 churned customers out of 7,043).

3.3 Train–Validation–Test Split

The data’s 7,043 records were split into 60-20-20 percent subsets to preserve the class distribution across all partitions. The result was a training set of n = 4,225 (60%), a validation set of n = 1,409 (20%), and a test set of n = 1,409 (20%). These subsets serve distinct purpose in the study. The training set was used for model fitting and cross-validation. The validation set was used for comparing models and selecting the final candidate. The test set was reserved exclusively for the final evaluation of the chosen model and was not used during any stage of model selection or hyperparameter tuning (Kuhn and Johnson, 2013). The three-way split of the data ensured that the test-set performance reflected a genuine out-of-sample generalization rather than optimistic estimates arising from information leakage.

3.4 Preprocessing Pipeline

All preprocessing steps were implemented within scikit-learn’s ColumnTransformer and Pipeline framework to ensure consistency across all data splits and to prevent data leakage. A median imputation was applied for numerical features (SeniorCitizen, tenure, MonthlyCharges, TotalCharges) which was then followed standard scaling. The choice of Median imputation over mean imputation was because the distribution of TotalCharges was highly right-skewed, therefore the median more robust to extreme values (Kuhn and Johnson, 2013). Standard scaling transforms features to zero mean and unit variance, which is crucial for Logistic Regression and for consistency of input ranges across all models.

The most frequent imputation and one hot encoding was applied to all categorical features, including gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, and PaperlessBilling, PaymentMethod. One-hot encoding creates separate binary columns for each category, avoiding the introduction of artificial ordinal relationships between categories. The encoder is configured to handle invisble categories during inference, preventing runtime errors when new data is encountered.

3.5 Handling Class Imbalance with SMOTE

The 26.5 per cent churn rate confirmed that the dataset is moderately imbalanced. To prevent models from favouring the majority class, SMOTE was applied to generate synthetic minority-class samples by interpolating between existing churn cases (Chawla et al., 2002). Critically, SMOTE was integrated into the imbalanced-learn Pipeline (ImbPipeline) and applied only within each training fold of the cross-validation procedure. This design ensures that synthetic samples never contaminate the validation or test data, preserving the integrity of performance estimates (Lemaitre et al., 2017).

3.6 Model Selection and Hyperparameter Tuning

Classification involved modelling four models representing different modelling techniques. The Logistic Regression served as an interpretable linear baseline with regularization controlled by the C parameter and optional class weighting. Decision Tree was fitted to model non-linear relationships through recursive partitioning, with depth, minimum samples per split, and minimum samples per leaf as key hyperparameters. The Random Forest, being an ensemble model, extended the performance of Decision Trees through bagging and random feature subsets, with additional hyperparameters for the number of trees, maximum features per split, and class weighting. XGBoost provided gradient-boosted trees with regularization parameters (alpha, lambda), learning rate, subsampling, and column sampling (Chen and Guestrin, 2016).

Hyperparameter tuning was performed using RandomizedSearchCV with 50 iterations per model and five-fold stratified cross-validation. The scoring metric was average precision (AUC-PR), which evaluated the area under the precision-recall curve. AUC-PR was chosen over raw recall since it penalizes both missed churners (low recall) and excessive false positives (low precision) simultaneously. This provides a more balanced optimization target for imbalanced classification (Saito and Rehmsmeier, 2015). The search distributions were chosen to cover broad, practically relevant ranges: uniform distributions for continuous parameters (e.g., C from 0.001 to 10, learning rate from 0.01 to 0.3), integer-uniform for discrete parameters (e.g., max_depth from 3 to 20), and explicit lists for categorical options (e.g., class_weight as None or 'balanced').

3.7 Evaluation Strategy

The classifiers were evaluated using the validation and the test data on six key metrics, including accuracy, precision, recall, F1-score, AUC-ROC, and AUC-PR. The validation-set results were used for model comparison and selection. The AUC-ROC was the primary selection criterion as it evaluated the discriminative power across all decision thresholds rather than at a single fixed cut-off (Fawcett, 2006). The test set was touched only once, for the final evaluation of the selected model. Confusion matrices were generated to examine the specific types of errors (false positives versus false negatives) produced by each model.

3.8 Explainability with SHAP

Model explainability was implemented using SHAP across all model families to ensure a consistent explanation framework. For tree-based models (Random Forest and XGBoost), TreeSHAP was used, which computes exact Shapley values in polynomial time (Lundberg et al., 2020). For Logistic Regression, LinearExplainer was used, which produces additive explanations consistent with the model's linear structure. SHAP summary plots and bar plots were generated to identify global feature importance patterns and to verify whether different model families agree on the key churn drivers.

3.9 Tools and Reproducibility

The entire analytical pipeline was implemented in latest Python 3.14 using established scientific computing libraries. Data manipulation and cleaning were performed using pandas (McKinney, 2010) and NumPy. Preprocessing, model training, and evaluation used scikit-learn (Pedregosa et al., 2011), with SMOTE provided by imbalanced-learn (Lemaitre et al., 2017). XGBoost was implemented using the xgboost library (Chen and Guestrin, 2016). SHAP analysis used the shap library (Lundberg and Lee, 2017). matplotib and seaborn were used to generate all the graphs and charts. Prior to creating and running cells, a fixed random seed (SEED = 42) was set in the first cell and used throughout to ensure full reproducibility of results. The complete codebase is available in a public GitHub repository (see Appendix A).

4. Exploratory Data Analysis (EDA)

4.1 Purpose

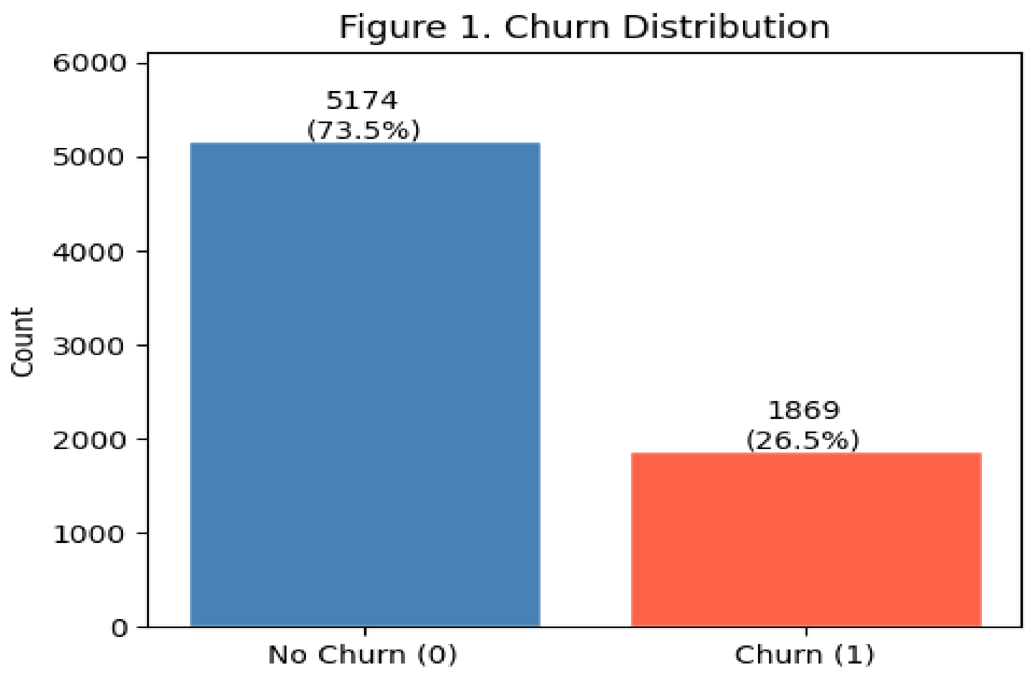
The EDA process was the preliminary analysis conducted to explore the structure and characteristics of the Telco Customer Churn dataset. The process sought to go beyond basic descriptives statistics, exploring the distributions of key variables, examine relationships between features and the churn outcome, informing the preprocessing and modelling decisions.

The EDA analysis also revealed patterns that had direct implications for model design and business interpretation.

4.2 Target Variable Distribution

The binary outcome variable, churn, comprised 5,174 customers (73.5%) that were retained, while the remaining 1,869 customers (26.5%) churned as illustrated in Figure 1. The distribution suggested moderate class imbalance, consistent with typical churn rates reported in telecommunications research (Burez and Van den Poel, 2009). Notably, the imbalance was not extreme enough to require aggressive resampling. However, the imbalance was sufficient to bias standard classifiers toward the majority class if left unaddressed. This observation led to the adoption of SMOTE during model training.

Figure 1:*Churn Distribution (0 = No Churn, 1 = Churn). Bar chart showing 5,174 retained customers (73.5%) and 1,869 churned customers (26.5%).*

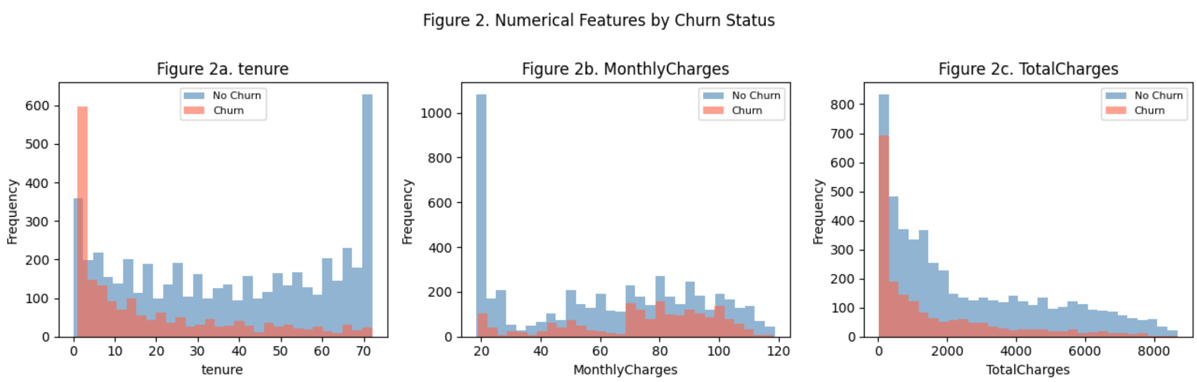


4.3 Numerical Feature Distributions

Histograms of the four numerical features, split by churn status, revealed several important patterns as shown in Figure 2. Tenure shows a bimodal distribution for the overall population, with peaks at both short and long tenure values. However, when disaggregated by churn status, customers who churned were concentrated in the 0-12th month range, while retained customers were even across the full 0-72nd month range. The finding indicates that customers are more likely to churn within their earlier months of associating with the service provider, which is consistent with earlier findings by Neslin et al. (2006).

The MonthlyCharges variable was significantly different between churn groups with churners having higher monthly charges (Mean = 74.4) than the customers who were retained (Mean = 61.3). This was an important insight suggesting that churn could be driven by higher monthly charges. The TotalCharges histogram showed a right-skewed distribution. Furthermore, the variable was strongly correlated with tenure, which was expected since cumulative spending increases with time. The SeniorCitizen variable is binary, with approximately 16 per cent of customers classified as senior citizens.

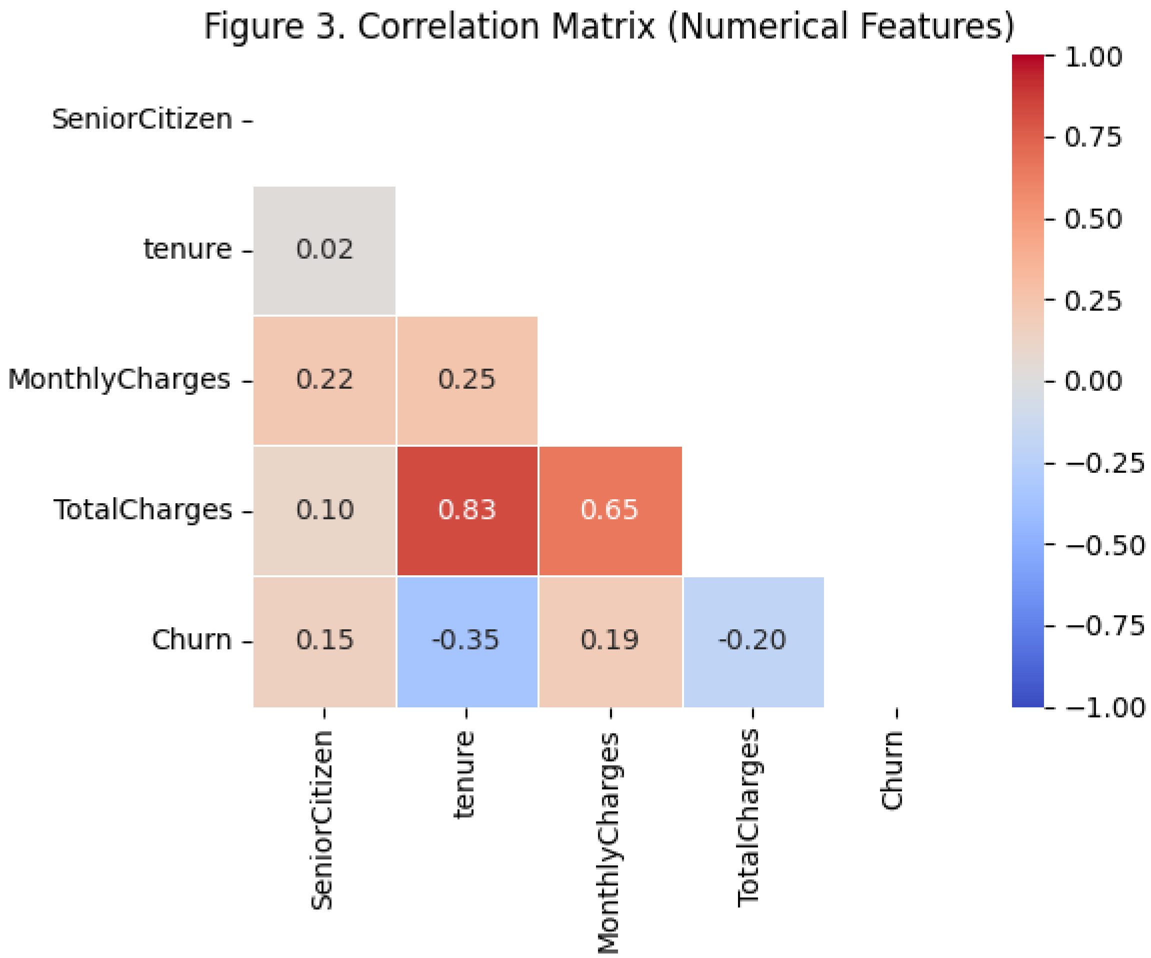
***Figure 2**: Numerical Feature Distributions by Churn Status. Histograms of tenure, MonthlyCharges, TotalCharges, and SeniorCitizen, coloured by churn label*



4.4 Correlation Analysis

Figure 3 presented the Pearson correlation matrix which validated several expected relationships. Tenure and TotalCharges exhibit a strong positive correlation (r = 0.83), reflecting the mechanical relationship between service duration and cumulative spending. Churn was negatively correlated with tenure (r = −0.35) and TotalCharges (r = −0.20), suggesting that customers with longer tenures, and high spenders were unlikely to churn. MonthlyCharges demonstrated a medium positive correlation with churn (r = 0.19), consistent with the hypothesis that higher monthly costs increase churn risk. These correlations inform feature selection and help validate the expected direction of effects in the models.

Figure 3: *Correlation Matrix of Numerical Features. Heatmap showing Pearson correlations among SeniorCitizen, tenure, MonthlyCharges, TotalCharges, and Churn.*

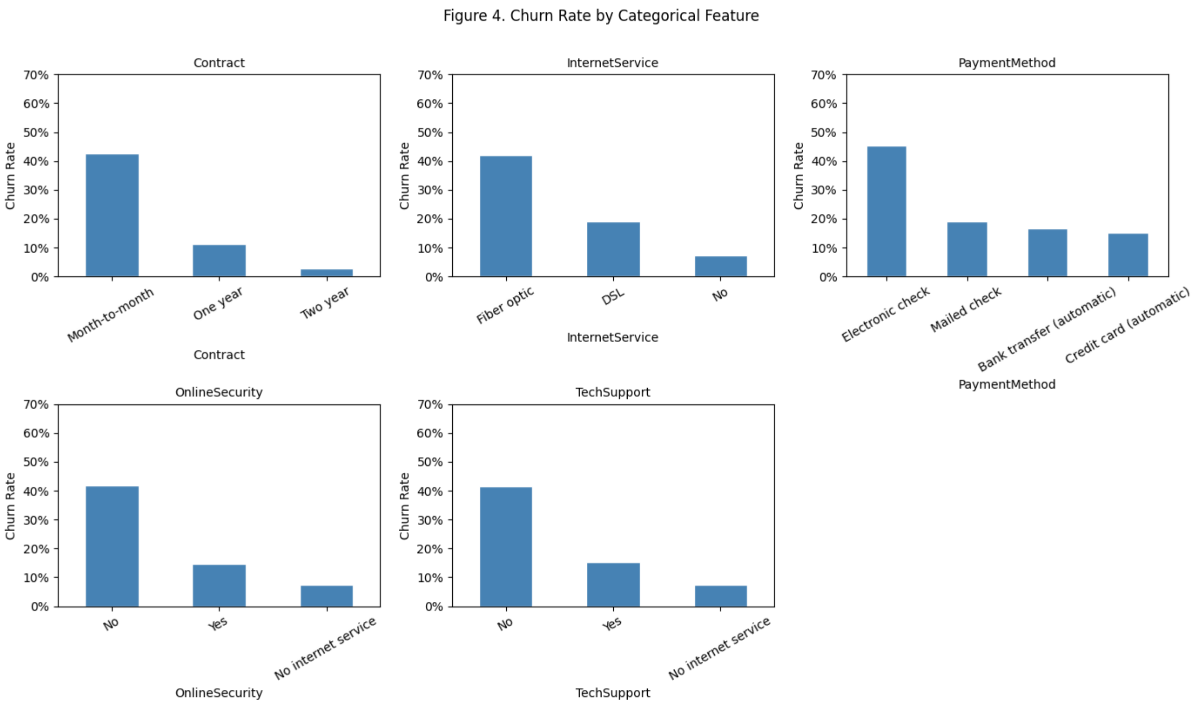


4.5 Categorical Feature Analysis

Figure presented the analysis of key categorical features against churn status revealing three important insights. First, the highest proportion of churners came from customers who were on month-to-month contracts. Approximately 4 in 10 customers on month-to-month contract were at churn risk. Customers on one-year contacts followed, with 11 out of every 100 being at churn risk. Two-year contracted customers had the least chances of churn, with only 3 in every 100 at risk of churning. This finding underscored the role of contractual commitment in retention and suggests that incentivizing contract upgrades could be an effective retention strategy.

The second observation relates to Internet service usage. Fiber optic internet customers churned at a higher rate than DSL customers, perhaps because of the higher monthly charges. This pattern suggested that the perceived value proposition of fiber optic service did not justify its price premium for some customer segments. Third, customers without online security or tech support services churned at roughly double the rate of those with these add-ons. This indicated that service engagement acts as a retention mechanism.

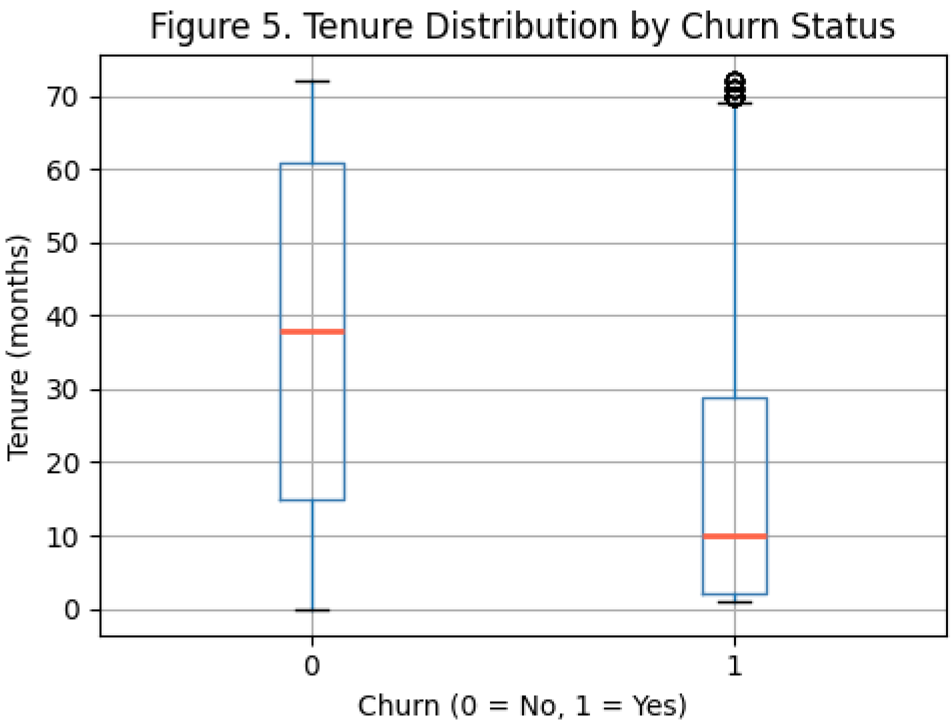
Figure 4: *Churn Rates by Contract Type, Internet Service, and Online Security. Grouped bar charts showing churn proportions for key categorical features.*



4.6 Tenure and Churn: Detailed Examination

The boxplot in Figure 5 presented the comparison of tenure by churn status. The figure showed that Churners had a median tenure of approximately 10 months, while retained customers had a median of approximately 38 months. The interquartile range for churners is almost entirely within the first two years of the customer lifecycle. This finding demonstrated that tenure was the single most discriminative feature. For businesses, initiating customer targeting programs in their first 10 months could significantly boost retention efforts.

Figure 5: *Tenure Distribution by Churn Status. Boxplot comparing tenure distributions for churned and retained customers.*



4.7 Missing Value Analysis

A systematic examination of missing values across all 20 features revealed that only the TotalCharges variable contained missing entries, with precisely 11 records affected. These missing values originated from blank string fields in the raw CSV file that produced NaN values upon numeric conversion. Further investigation showed that all 11 records correspond to customers with zero or near-zero tenure—new customers who had not yet generated any billing history. Since TotalCharges is a function of tenure and monthly charges, the absence of accumulated charges for these new customers is logically consistent rather than indicative of a data quality problem. Given the small number of affected records (0.16% of the dataset) and the importance of retaining these new customers for churn analysis—who are precisely the high-risk group identified by the tenure analysis—median imputation was chosen as the appropriate handling strategy, to be applied within the preprocessing pipeline to prevent data leakage.

4.8 Service Subscription Patterns

A deeper examination of service subscription patterns revealed that customer engagement with value-added services varies considerably and has meaningful implications for churn. Approximately 90 per cent of customers subscribe to phone service, and fibre optic is the most common internet service type (3,096 customers), followed by DSL (2,421) and no internet service (1,526). Among customers with internet service, a substantial proportion do not subscribe to additional protective services: 3,498 customers lack online security, 3,088 lack online backup, 3,095 lack device protection, and 3,473 lack tech support. This pattern of low engagement with add-on services is particularly concerning because, as the churn analysis shows, customers without these services churn at roughly double the rate of those with them. The implication for modelling is that the one-hot encoded versions of these service variables are likely to carry significant predictive information about churn risk.

4.9 Payment Method and Billing Analysis

The analysis of payment methods revealed that electronic check is the most common payment method (2,365 customers), followed by mailed check, bank transfer, and credit card. Notably, electronic check users exhibit the highest churn rates among all payment methods, which may be attributable to the manual nature of this payment process—each payment requires active effort from the customer, creating a recurring decision point at which churn can occur. By contrast, automatic payment methods (bank transfer and credit card) show lower churn rates, suggesting that payment friction is a meaningful, though indirect, contributor to churn behaviour. Additionally, 4,171 customers (59%) use paperless billing, and this group shows a moderately higher churn rate than those receiving paper bills, possibly because paperless billing customers are more digitally engaged and therefore more exposed to competitor offerings.

4.10 Key Insights from EDA

The exploratory analysis yielded six principal insights that directly informed the modelling strategy. First, the dataset is moderately imbalanced (26.5% churn), requiring SMOTE during training. Second, short tenure is the strongest individual predictor of churn, with churners concentrated in the first 12 months. Third, higher monthly charges are associated with increased churn risk, suggesting price sensitivity. Fourth, month-to-month contracts are a major churn driver, with a 43% churn rate compared to 3% for two-year contracts. Fifth, service engagement—particularly online security and tech support—is associated with substantially lower churn rates. Sixth, payment method influences churn, with electronic check users showing the highest churn rates. These findings established the analytical foundation for the preprocessing pipeline and the interpretation of model results.

5. Implementation of Algorithms and Models

5.1 Overview

Prediction of churn in this study involved implementing four machine learning classifiers, namely the Logistic Regression (linear), Decision Tree (single non-linear model), Random Forest (bagged ensemble), and XGBoost (boosted ensemble). All models were trained using identical preprocessing pipelines and evaluated on the same metrics to ensure that interpretation and comparison was consistent and fair. Before the classification results are discussed, the succeeding section discusses the model configuration for each of the four classifiers, the hyperparameter search spaces, the tuning results, and the validation-set comparison used for model selection.

5.2 Logistic Regression

Logistic Regression estimates the probability of churn as a logistic function of a linear combination of input features (Hosmer et al., 2013). Despite its simplicity, it remains one of the most widely used classifiers in churn prediction due to its interpretability and competitive performance on well-preprocessed data. The hyperparameter search space included the regularization strength C, sampled from a continuous uniform distribution over [0.001, 10], and the class_weight parameter, tested as either None or ‘balanced’. The ‘balanced’ alternative adjusted class weights inversely proportional to class frequency, providing an additional mechanism for handling imbalance beyond SMOTE.

The optimal configuration, identified through randomized search, used a very low regularization strength (C = 0.009) with balanced class weights, achieving a cross-validated AUC-PR of 0.657. The strong regularization suggests that the classifier benefited from the constraining the coefficients, which prevents overfitting to noise in the training data.

5.3 Decision Tree

The Decision Tree classifier recursively partitions the feature space into regions that maximize class separation (Breiman et al., 1984). Key hyperparameters control the complexity of the tree: max_depth limits the number of sequential splits, min_samples_split sets the minimum number of samples required to split an internal node, and min_samples_leaf sets the minimum number of samples at each terminal node. The search space covered max_depth from 3 to 19, min_samples_split from 2 to 49, min_samples_leaf from 1 to 29, and class_weight as None or 'balanced'.

The optimal Decision Tree configuration for the classifier was characterized by a maximum depth of 10, minimum 12 samples per split, and minimum 24 samples per leaf, with balanced class weights. The classifier yielded a cross-validated AUC-PR of 0.594, which was the lowest compared to the other three models. The relatively shallow depth and high leaf constraints indicated that the pruned tree avoids overfitting but at the cost of expressive power.

5.4 Random Forest

The Random Forest being an ensemble model builds on the performance of the Decision Tree model through training an ensemble of trees on bootstrap samples of the data. During the training, each tree considers only a random feature at each split (Breiman, 2001). This process, combining bagging and random feature selection, leads to variance reduction and boosts the model's generalization. The hyperparameter search space for this classifier included the number of trees (100 to 499), max_depth (None or 5 to 24), min_samples_split (2 to 19), min_samples_leaf (1 to 14), max_features ('sqrt', 'log2', or 0.5), and class_weight (None or 'balanced').

The optimal Random Forest configuration utilized 151 trees with a maximum depth of 12, ‘sqrt’ feature selection, minimum 18 samples per split, and minimum 14 samples per leaf, without class weighting. The configuration led to a cross-validated AUC-PR of 0.655, closely matching Logistic Regression. The classifier’s moderate depth and relatively high leaf constraints suggest a preference for smoother decision boundaries, consistent with the bias-variance trade-off in ensemble methods.

5.5 XGBoost

The fourth classifier, XGBoost, implements gradient-boosted decision trees with built-in L1 and L2 regularization, which helps prevent overfitting (Chen and Guestrin, 2016). Among all the four classifiers, the XGBoost had the most broad hyperparameter search space that covered the number of boosting rounds (100 to 499), max_depth (3 to 11), learning_rate (0.01 to 0.3), subsample (0.6 to 1.0), colsample_bytree (0.5 to 1.0), reg_alpha (0 to 1), reg_lambda (0.5 to 2.5), and scale_pos_weight (1 or 2.77, where 2.77 approximates the negative-to-positive class ratio).

The optimal XGBoost configuration utilized 195 trees with a depth of 4, learning rate of 0.057, subsample of 0.974, colsample_bytree of 0.587, moderate L1 regularization (alpha = 0.549), strong L2 regularization (lambda = 1.929), and scale_pos_weight of 2.77. The configuration also attained the highest cross-validated AUC-PR of 0.662. however, the model configuration’s shallow depth and strong regularization implied that the model preferred many weak learners over a few deep trees, which is consistent with best practice for gradient boosting (Friedman, 2001).

5.6 Validation-Set Comparison

The validation-set performances for the four classifiers are presented in Table I below. The models were evaluated at the default 0.5 decision threshold.

Table 1*: Validation-set Performance Comparison*

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.753	0.523	0.805	0.634	0.854
Decision Tree	0.765	0.547	0.674	0.604	0.817
Random Forest	0.790	0.582	0.738	0.651	0.857
XGBoost	0.732	0.497	0.840	0.624	0.853

As indicated in the table, the Random Forest attained the highest validation AUC-ROC (0.857), followed closely by the Logistic Regression model (0.854), XGBoost (0.853), and finally the Decision Tree which ranked fourth (0.817). A further exploration of individual classifier metrics, the XGBoost achieved the highest recall (0.840) and also the lowest precision (0.497). The contrast reflects an aggressive churn prediction model with many false positives. The Random Forest provided the most balanced trade-off between precision (0.582) and recall (0.738), reflected in its highest F1-score (0.651). Logistic Regression achieved high recall (0.805) with moderate precision (0.523).

The comparison chart in Figure 6 and the ROC curves presented in Figure 7 give a visual summary of the exhibited individual model characteristics. The ROC curves demonstrate that the four classifiers attained sensible discrimination, with the three ensemble/linear models closely clustered and the Decision Tree trailing. The precision-recall curves presented in Figure 8 provide further evidence of the Random Forest's strength in balancing the recall-precision trade-off.

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Figure 6: *Model Comparison Bar Chart — Validation and Test Metrics. Grouped bar chart comparing accuracy, precision, recall, F1, and AUC-ROC across all four models on both validation and test sets*

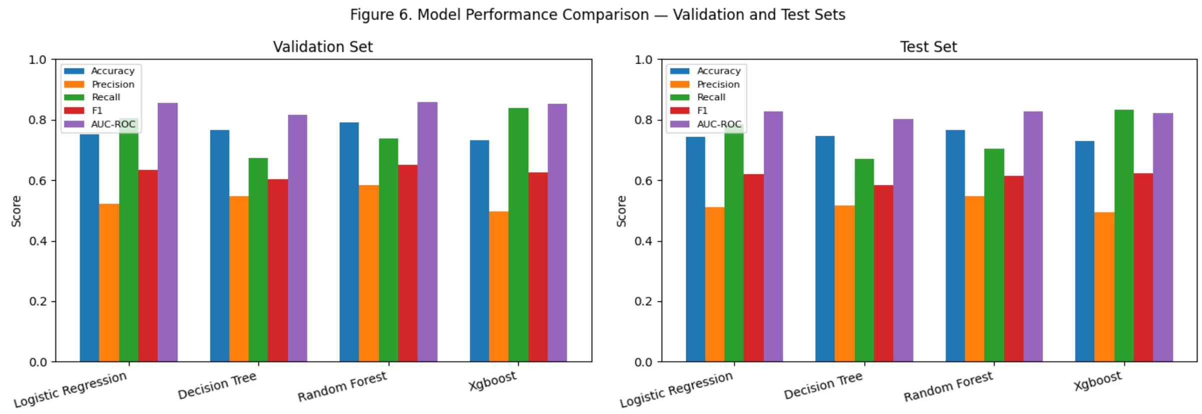


Figure 7: *ROC Curves — Validation and Test Sets. ROC curves for all four models on validation (left) and test (right) sets.*

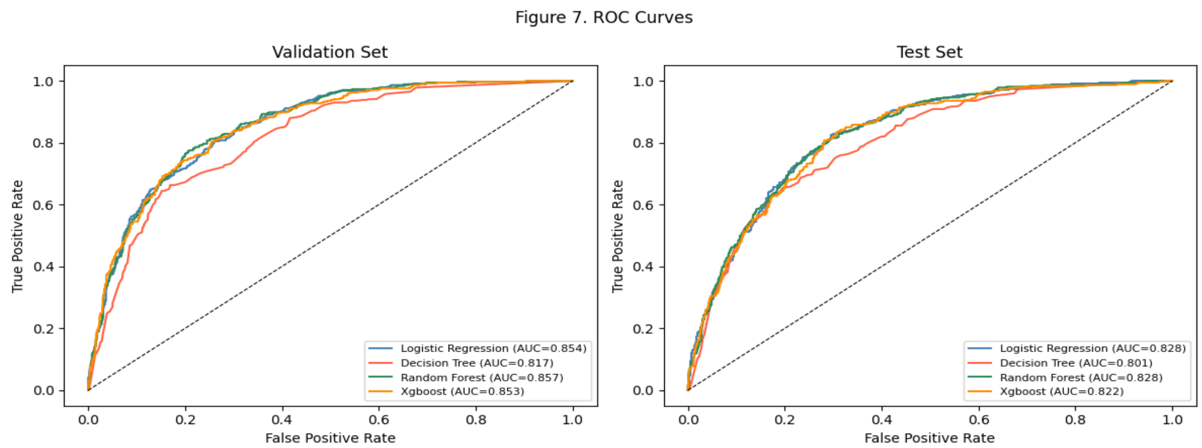
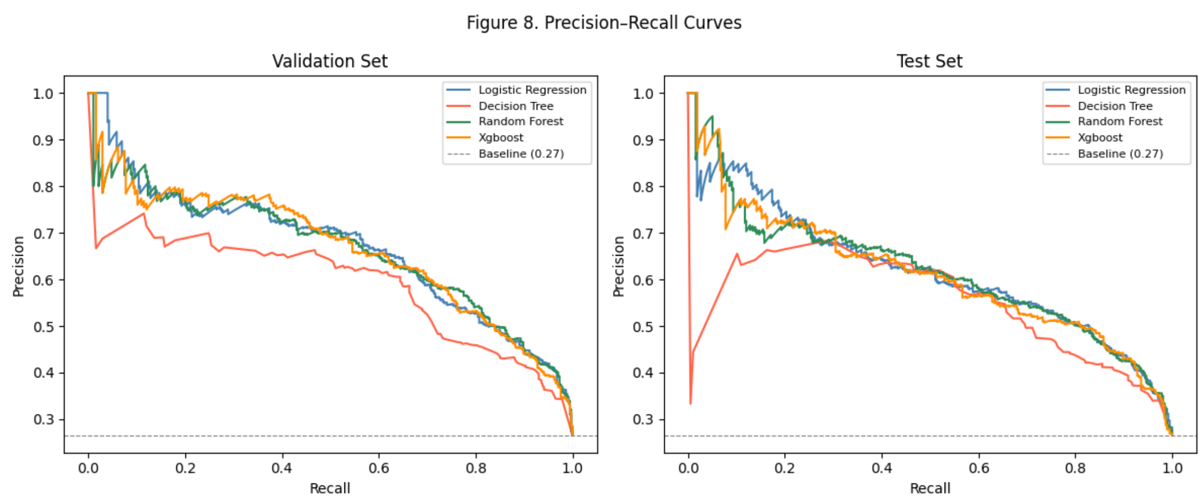


Figure 8: *Precision-Recall Curves — Validation and Test Sets. PR curves for all four models on validation (left) and test (right) sets.*



5.7 Why the Models Differ

As discussed in the literature review, the four models are unique in their own ways and are applicable depending on the data and the specific objectives at hand. Therefore, the performance disparities inform about the nature of the churn prediction task. The Decision Tree with its relatively weak performance (validation AUC-ROC of 0.817) underscores the fundamental limitation of single-tree models. The models are prone to high variance and can only approximate complex decision boundaries through axis-aligned splits. The tree's best configuration required heavy pruning (max_depth = 10, min_samples_leaf = 24) to avoid overfitting, which limits its expressive capacity.

The surprise strong performance by the Logistic Regression (AUC-ROC 0.854), implies that the relationship between features and churn is significantly linear after one-hot encoding, which effectively linearizes the categorical variables. The balanced class weights and strong regularization (C = 0.009) indicated that the model benefited from constraining coefficients. This was aligned with the relatively large number of one-hot encoded features (45 after encoding) compared to the training set size.

Random Forest and XGBoost achieved similar overall AUC-ROC (0.857 and 0.853 respectively), however, they were differentiated based on precision-recall trade-offs. The XGBoost's higher recall (0.840) at the expense of lower precision (0.497) highlighted the classifier's scale_pos_weight parameter being set to 2.77, which aggressively weighted the minority class. Random Forest achieved a more balanced profile because its optimal configuration did not use class weighting, relying instead on the ensemble's natural ability to smooth decision boundaries.

5.8 Final Model Selection

Random Forest emerged as the best and optimal classifier following its high AUC-ROC (0.857) than the other three classifiers. The rationale for using AUC-ROC as the selection criterion, rather than a single metric such as recall, is that AUC-ROC evaluates the model's discriminative ability across all possible decision thresholds. This makes it a more robust and generalizable criterion than optimizing for a single threshold-dependent metric (Fawcett, 2006). If the business imposes a specific cost constraint—for example, that no more than 40 per cent of flagged customers should be false positives—the decision threshold can be adjusted post hoc on the selected model to meet that constraint.

6. Results and Analysis

6.1 Test-Set Performance

Table 2 presents the test-set results. The held-out test set was used exclusively for the final evaluation of the selected model, Random Forest, and for reporting the performance of all four models for completeness.

Table 2: *Test-set Performance Comparison*

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.743	0.510	0.786	0.619	0.828
Decision Tree	0.746	0.517	0.671	0.584	0.801
Random Forest	0.767	0.547	0.703	0.615	0.828
XGBoost	0.730	0.495	0.834	0.622	0.822

Random Forest achieved a test AUC-ROC of 0.828 that closely matched the Logistic Regression and marginally exceeded the XGBoost (0.822) and Decision Tree (0.801). The figures suggested reasonable generalization given the small gap between validation (0.857) and test (0.828) AUC-ROC. However, the drop suggested some degree of optimism in the validation estimates. Notably, the ranking of models was preserved between validation and test sets, confirming the reliability of the model selection process.

Random Forest's test recall of 0.703 translates to approximately 70 per cent of actual churners are correctly identified. Additionally, the model's precision of 0.547 implies for the 54.7% of the customers who were predicted to churn, actually churned. In the real business world, the metric means that if a retention program targets 100 customers, 55 of them would be at genuine risk of churning. Whether this false positive rate is acceptable will depend on the cost of the retention intervention in comparison to the cost of losing a customer.

6.2 Stability and Generalization

Table 3 compared validation and test performance for the selected model, Random Forest, to examine stability.

Table 3: *Random Forest — Validation vs. Test Stability*

Metric	Validation	Test	Difference
Accuracy	0.790	0.767	−0.023
Precision	0.582	0.547	−0.035
Recall	0.738	0.703	−0.035
F1	0.651	0.615	−0.036
AUC-ROC	0.857	0.828	−0.029

The table shows that all metrics had modest declines of 2 to 4 percentage points from validation to test. The decline was within the expected range for a well-regularized model on a dataset of this size. The consistency across metrics suggested that the model generalized reliably rather than overfitting to the validation set.

6.3 Confusion Matrix Analysis

The confusion matrices presented Figure 9 revealed that the distribution of prediction errors. On the test set, Random Forest produced 263 true positives (correctly classified churners), 112 false positives (retained customers misclassified), 219 true negatives correctly classified, and 815 true negatives. The 112 false negatives represented churners who were missed by the model and would not receive retention interventions. In the real business world, the false positives represent the cost of model imperfection. This number can, however, adjusted by reduction at the expense of increased false positives.

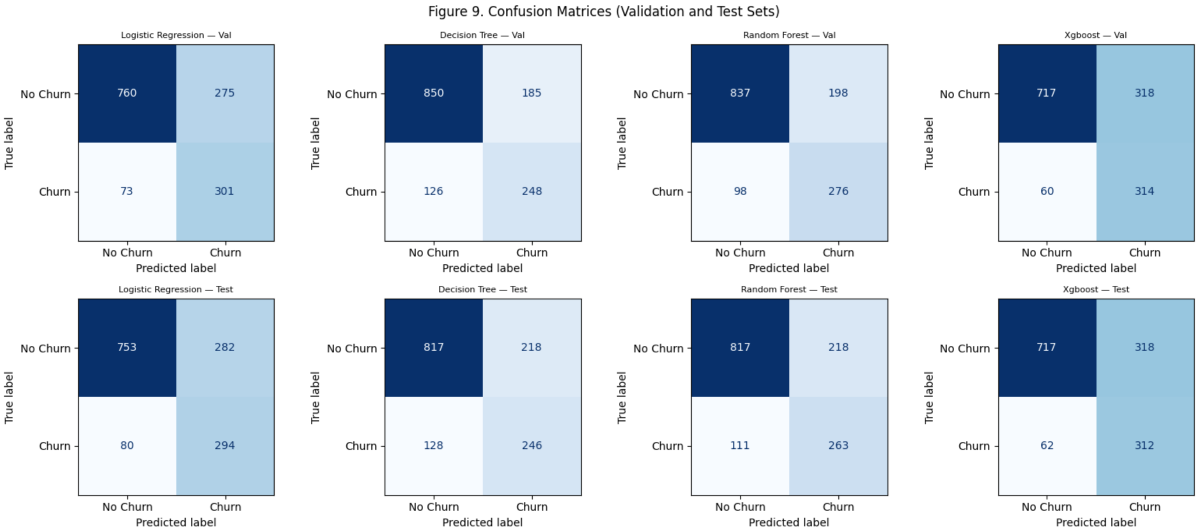
6.4 Cross-Model Agreement

The study results point to a consensus across the model families on overall performance ranking. The four classifiers configured in this study attained test AUC-ROC values that ranged from 0.801 to 0.828. This implied that the dataset's predictive ceiling is approximately 0.83 for the available feature set. The convergence was consistent with the findings of Grinsztajn et al. (2022), who found that on moderate-sized tabular datasets, the performance disparity between linear models and tree-based ensembles is often small. The overarching take from the study was that, sophisticated modelling algorithms were not necessary, but new data sources, including customer satisfaction surveys, complaint logs, or competitor pricing.

6.5 Threshold Sensitivity

The metrics in Tables 1 and 2 were reported at the default decision threshold of 0.5. However, for businesses, the optimal threshold is usually based on the relative costs of false positives, unnecessary retention spending, and false negatives—lost customers. The precision-recall curves indicated that for Random Forest, increasing the threshold from 0.5 to 0.6 would increase precision from 0.547 to approximately 0.62 while reducing recall from 0.703 to approximately 0.58. On the contrary, lowering the threshold to 0.4 would increase recall to approximately 0.80 but reduce precision to approximately 0.45. In practice, a telco business that deploys low-cost customer retention strategies such as sending promotional emails, might prefer the lower threshold. In contrast, a business that deploys high-cost retention strategies such as equipment upgrades for customers might prefer the higher threshold. The choice of AUC-ROC as the model selection criterion ensures that the selected model performs well across this full range of operating points.

Figure 9: ***Confusion Matrices for All Four Models — Validation (left) and Test (right) sets.***



7. Model Explainability

7.1 SHAP Analysis Overview

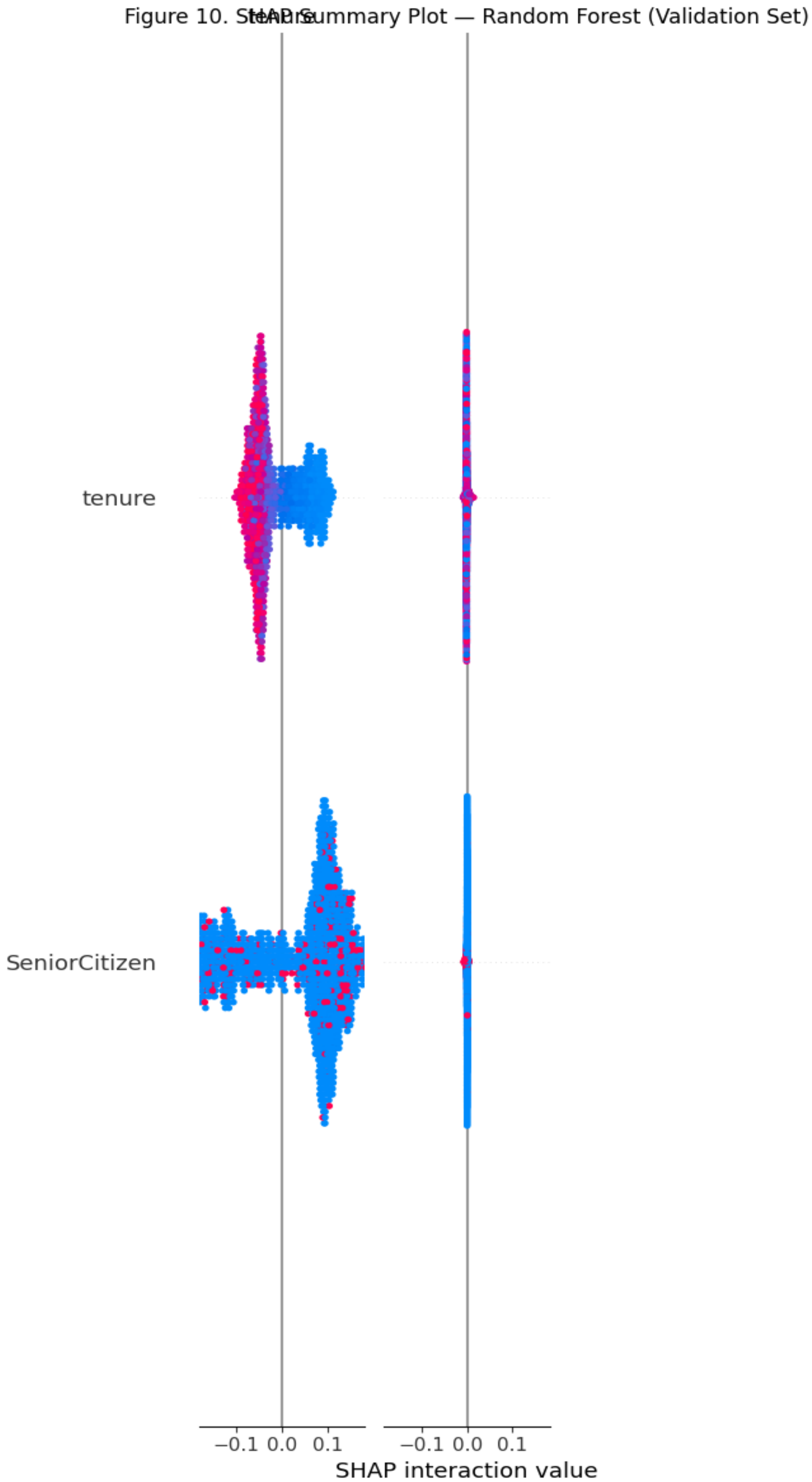
To ensure transparency and to generate actionable insights, SHAP analysis was applied to all three primary model families: Random Forest, XGBoost, and Logistic Regression. Using SHAP across all models provides a unified explanation framework that allows direct comparison of feature importance rankings, increasing confidence in the findings when different model families agree on the key drivers (Lundberg and Lee, 2017).

7.2 Random Forest SHAP Summary

Figure 10 presents the SHAP summary plot for Random Forest ranking feature importance by mean absolute SHAP value, with individual data points colored by feature value. The plot indicates that tenure was the most significant feature with low tenure values, red dots on the right side of the plot, pushing predictions strongly toward churn. conversely, high tenure values push predictions toward retention. MonthlyCharges and TotalCharges features emerged second and third, respectively. An increase in the monthly charges corresponded with an increase in churn probability, while higher total charges which reflected a customer's lifetime value decreased churn likelihood.

The one-hot encoded Contract_Month-to-month feature ranked among the top five. This indicated that month-to-month contracts were a strong positive determinant of churn. InternetService_Fiber optic also appears prominently, consistent with the EDA finding that fibre optic customers had relatively higher churn rates compared to DSL or customers without internet service. OnlineSecurity_No and TechSupport_No contribute positively to churn predictions, supporting the insight that customers without these protective services are at greater risk.

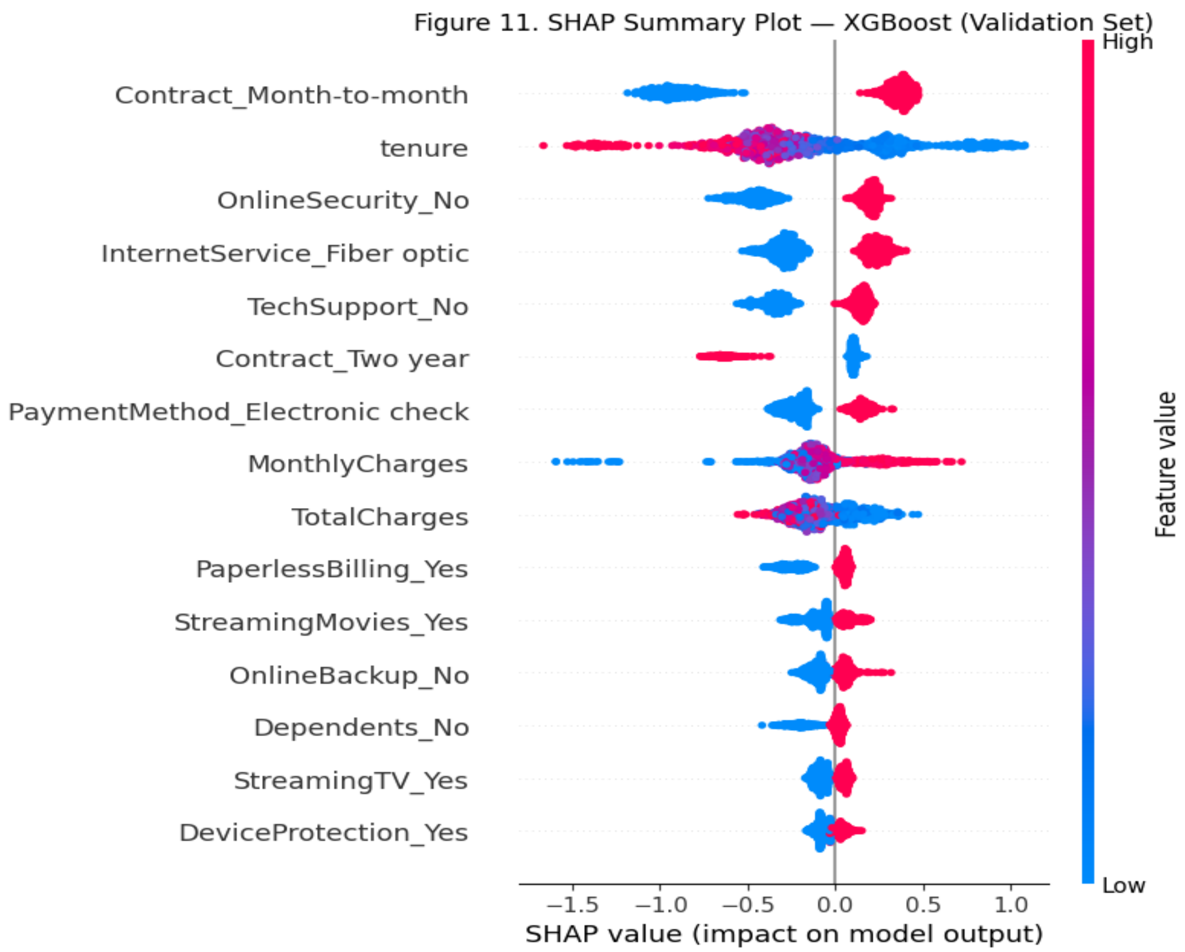
Figure 10: *SHAP Summary Plot — Random Forest. Beeswarm plot showing feature contributions to churn predictions, coloured by feature value.*



7.3 XGBoost SHAP Summary

Figure 11 presented the SHAP summary plot for XGBoost showing a highly consistent ranking with Random Forest. Tenure, Contract_Month-to-month, MonthlyCharges, and TotalCharges were also the major determinants of churn. This is was similar to the Random Forest classifier. The consistent pattern exhibited by the two tree-based models strengthens confidence in these features as significant determinants of churn rather than being considered artefacts of an algorithm. However, there were minor differences in ranking, for instance, XGBoost placed slightly more importance on InternetService_Fiber optic, reflecting the models' different inductive biases. Nevertheless, these differences do not change the overall conclusions.

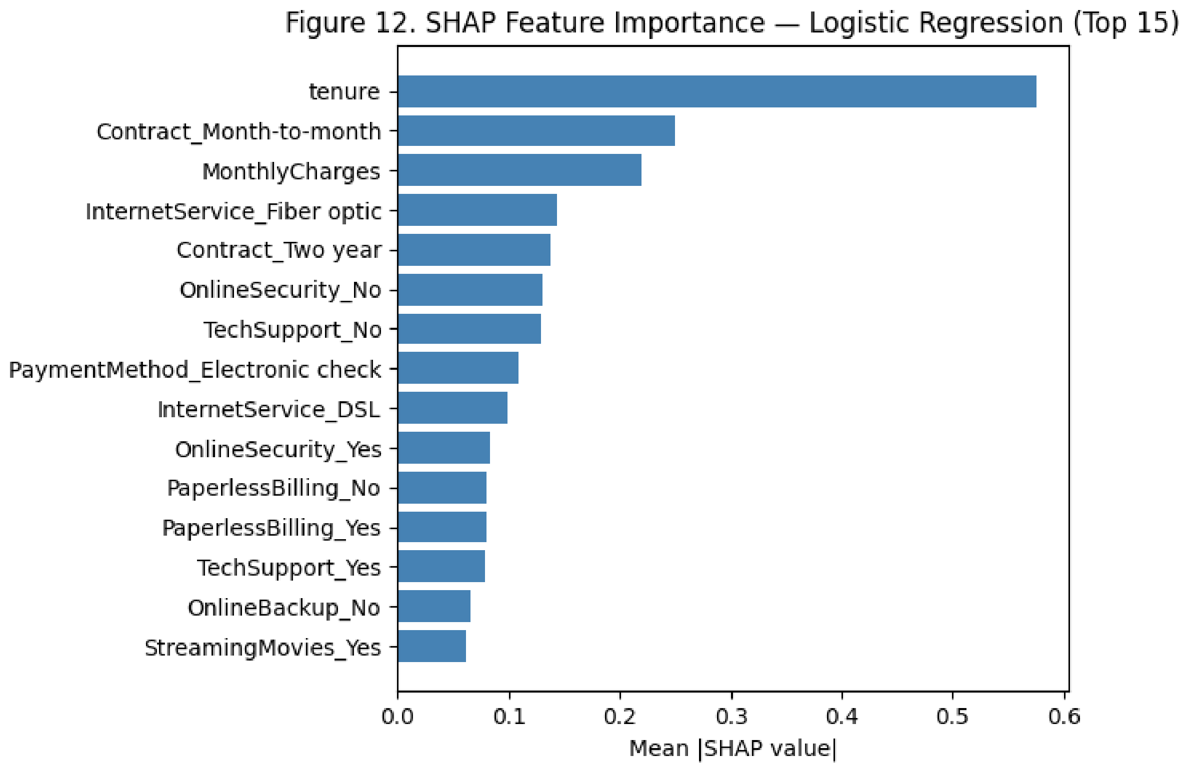
Figure 11: *SHAP Summary Plot — XGBoost. Beeswarm plot showing feature contributions to churn predictions.*



7.4 Logistic Regression SHAP Analysis

The computation of SHAP values for the logistic model differed from the other methods. It involved use of the LinearExplainer producing additive explanations consistent with the model's linear structure. Figure 12 shows the top 15 features by mean absolute SHAP value. Just like in the previous models, tenure, MonthlyCharges, Contract_Month-to-month, and TotalCharges emerged as the most significant determinants of churn. The consistency between the linear SHAP explanations and the tree-based SHAP plots provides strong cross-validation of the feature importance findings and increases confidence that these relationships are genuine rather than model-specific artefacts.

Figure 12: *SHAP Feature Importance — Logistic Regression (Top 15). Horizontal bar chart of mean absolute SHAP values.*



7.5 Interpretation of Key Features

The three models: Random Forest, XGBoost, and Logistic regression, converge on the top determinants of churn. Tenure, Monthly charges, and total charges emerge as the top three features, respectively. Tenure was consistently the single strongest predictor. Customers with short tenure, typically 10-12 months, were the most at risk of churn, consistent with the retention literature on early customer lifecycle vulnerability (Neslin et al., 2006). Monthly charges contribute positively to churn predictions, with high monthly charges driving away customers. Short contract length, specifically, Month-to-month contracts were a strong positive predictor of churn, reflecting the low switching costs associated with this contract type. Internet service type, specifically fibre optic, is associated with higher churn, possibly due to higher pricing without commensurate perceived value. Additionally, the absence of online security and tech support services increased churn risk, indicating that service bundling and engagement acted as retention strategies.

The consistency of SHAP rankings across model families was a particularly noteworthy finding. When a linear model (Logistic Regression), a bagged ensemble (Random Forest), and a boosted ensemble (XGBoost) all identify the same set of features as the most important churn drivers, the result is far more credible than a single-model analysis. This cross-model consensus reduces the risk that the identified drivers are artefacts of a particular algorithm’s inductive bias. it also boosts confidence that the identified features reflect genuine patterns in the underlying data. For business stakeholders, this convergence provides a stronger evidence base for investment in retention programmes targeting these specific drivers.

It is also worth noting the features that do not appear prominently in the SHAP rankings. Gender and Partner status contribute minimal SHAP values across all models, suggesting that demographic characteristics alone are poor predictors of churn in this dataset. This finding was consistent with the broader literature, which generally found that behavioral variables, what customers do, are more predictive than demographic variables, who customers are, in subscription-based services (Verbeke et al., 2012).

8. Business Insights and Recommendations

8.1 From Model to Decisions

The value of developing churn prediction models is not to obtain statistical accuracy alone; it should also lead to insightful recommendations for customer retention. In this paper, the SHAP analyses gives a pathway for connecting the study outputs and the business strategies by identifying features from each classifier that determine churn for different customer stratifications. The sections under this chapter translates the analytical findings into six actionable recommendations, each grounded in a specific churn driver identified through the modelling process.

8.2 Retention Strategy Recommendations

Table 4 summarizes the key churn drivers and corresponding retention recommendations.

Table 4: *Churn Drivers and Retention Recommendations*

Churn Driver	Recommended Action
Short tenure (< 12 months)	Introduce an onboarding loyalty discount for the first 6 months to reduce early-lifecycle churn. Research suggests that structured onboarding programmes can reduce early churn by 15–25 per cent (Reichheld and Schefter, 2000).
Month-to-month contract	Incentivise upgrades to one- or two-year contracts through price lock-in guarantees and loyalty rewards. The EDA shows that churn drops from 43% (month-to-month) to 3% (two-year), representing a 40-percentage-point reduction in churn risk.
High monthly charges	Offer tailored service bundles that reduce the perceived cost per service. Proactive value communication—such as usage reports showing the services a customer uses and their individual cost—may reduce price-driven churn.
Fibre optic without security add-ons	Bundle online security and device protection at a discounted rate with fibre optic plans. This simultaneously addresses the higher churn rate among fibre optic customers and the protective effect of security add-ons.

Electronic check payment	Encourage customers to switch to auto-pay methods (credit card or bank transfer) by offering a small monthly credit. Electronic check users may churn at higher rates due to the friction of manual payment processing.
No tech support / online security	Proactively offer a free trial of tech support and online security services to high-risk customers identified by the model. The SHAP analysis shows that the absence of these services is a consistent churn driver across all models.

8.3 Practical Implications

The model's predictions can be operationalized in several ways. The model developed in this study have the potential a monthly list of customers at risk of churn together churn probability. This is important for customer retention departments as they can use the probabilities and prioritize their outreach strategies. Secondly, the SHAP values for individual customers can be used to personalize the retention message: a customer whose churn risk is driven primarily by price might receive a discount offer, while a customer whose risk is driven by lack of service engagement might receive a free trial of additional services.

Businesses may adjust as appropriate the threshold for marking a customer as at-risk based on their economics. If the cost of a retention strategy, such as a discount or free trial, is low relative to the customer's lifetime value, a lower threshold—flagging more customers and accepting more false positives—is appropriate. However, if the retention strategy is expensive, a higher threshold that prioritizes precision may be preferred. This flexibility is one of the advantages of using AUC-ROC as the selection criterion, as it ensures the model performs well across a range of thresholds rather than being optimized for a single cut-off.

8.4 Cost-Benefit Framework

Consider the simplified cost-benefit analysis that illustrates the economic value of the churn prediction model. Let us assume that a telco company has an average customer lifetime value of $1,500, cost of retaining a customer is $60, and the probability that a strategy successfully prevents churn is 30 per cent (Verbeke et al., 2012). Under the described conditions, the expected benefit of intervening on a true positive is 0.30 * $1,500 = $450, while the cost is $60, resulting in a net benefit of $390 per true positive. The cost of a false positive, intervening on a customer who would not have churned, is $60 with no corresponding benefit. Using Random Forest's test-set performance, out of 1,409 test customers, the model correctly identifies 263 churners (true positives) and incorrectly flags 218 non-churners (false positives). We can, therefore, determine the estimated net benefit as: 263 * $390) − (218 * $60) = $102,570 − $15,780 = $86,790, per cycle. The above figures are hypothetical, however, they demonstrate that even a model with moderate precision can generate significant economic value when the cost of intervention is low relative to the customer lifetime value.

8.4 Limitations from a Business Perspective

While this study developed successfully developed four models to predict churn, it was limited in several ways. First, the study utilized a static snapshot dataset that did not capture temporal dynamics such as seasonal churn patterns or the effect of competitor promotions. The developed models have the assumption that past behavior is predictive of future behavior. This may not be the case in the practical business world, especially in the face of market disruptions. Secondly, the selected dataset did not have customer satisfaction scores, complaint history, or network quality metrics, factors which are likely to influence churn decisions in practice. Third, while the SHAP analysis identifies statistical associations between features and churn, it does not establish causal relationships. Correlation between fibre optic service and higher churn, for example, may be confounded by pricing or geographic factors not captured in the data.

8.5 Future Work

Based on the identified limitations for the study, several recommendations are suggested for future research. First, future researchers should prioritize other data sources beyond the Telco Customer Churn dataset used in this study. Data sources such as customer support interactions, network quality metrics, and competitor pricing would likely improve predictive performance. Second, threshold optimization based on explicit business cost ratios would enable more economically rational deployment decisions. Third, monitoring model performance over time and implementing automated retraining pipelines would address the risk of concept drift.

9. Conclusion

9.1 Summary

The study developed four end-to-end customer churn prediction models utilizing the publicly available Telco Customer Churn dataset. The models were trained using a structured preprocessing pipeline that incorporated median imputation, one-hot encoding, standard scaling, and SMOTE for class imbalance handling. Hyperparameters were tuned using RandomizedSearchCV with 50 iterations and five-fold stratified cross-validation, scored on AUC-PR to balance recall and precision. Among the four model, the Random Forest emerged as the best and optimal model recording the highest validation AUC-ROC (0.857) and also achieved a test AUC-ROC of 0.828, confirming strong generalization.

SHAP analysis also revealed that the most significant determinants of churn were tenure, monthly charges, and total charges, respectively. This was true for the Random Forest, XGBoost, and the Logistic Regression models. This study was able to summarize the findings into six evidence-based retention recommendations, ranging from onboarding loyalty programmes to service bundling strategies. This study provides empirical evidence that machine learning methodologies combined with rigorous evaluation and transparent.

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Appendices

Appendix A: GitHub Repository

The complete Python Jupiter notebook that can be used to reproduce the results with the chosen dataset can be assessed using the following GitHub repository:

https://github.com/ADHAOUI171/Customer_churn_Prediction_.git

Appendix B: AI Usage Declaration

ChatGPT was used in the project in a limited supportive capacity. First, it was used to assist with debugging Python code and troubleshooting technical issues. Second, it was used to generate the essay outline. All substantive analysis, interpretation, final writing, and academic judgments in the paper were solely my responsibility as the author.

The prompts used to debug the code are:

Prompt 1: “in VSCODE and getting a TypeError when using SMOTE inside a scikit-learn pipeline. The code I’m using is Pipeline ([('preprocessor', preprocessor), ('smote', SMOTE()), ('clf', LogisticRegression())]). The error I get tells me SMOTE is not a valid transformer. Please help me fix this?”

Prompt 2: “I'm trying to run SHAP on my Logistic Regression model but getting a shape mismatch error. My pipeline has a preprocessor and SMOTE step before the classifier. How do I extract the transformed validation data and pass it to shap.LinearExplainer with the correct feature names after one-hot encoding?”

Prompt 3: “TotalCharges column is showing as dtype object even though it should be numeric. When I try df['TotalCharges'].astype(float) I get a ValueError: could not convert string to float. I think some of the roes in my data have blank spaces instead of numbers. How can I convert this column and handle the blanks?”

The prompts can be accessed using the link https://chatgpt.com/share/69f26df1-a08c-83ea-ab89-42e7c6e0df52

Appendix C: Declaration of Originality

DECLARATION

This thesis is a product of my own work and is the result of nothing done in collaboration.

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